

Integration Theory for Multivariable Functions
Lecture 9 for Multivariable Calculus

Contents

Lecture Plan **1**

1 Review of Double Integrals from Lecture 8 **2**

1.1 Meaning of a Double Integral 2

1.2 Computing on Rectangles 2

1.3 Type I and Type II Regions 3

1.4 Changing the Order of Integration 4

1.5 Polar Coordinates for Double Integrals 5

1.6 Double-Integral Technique Checklist 5

2 Triple Integrals and Coordinate Systems **6**

2.1 Meaning of a Triple Integral 6

2.2 Rectangular Boxes and Fubini’s Theorem 6

2.3 Reduction to a Double Integral and a Single Integral 6

2.4 Cartesian Bounds for Common Solids 8

2.5 Cylindrical Coordinates 8

2.6 Spherical Coordinates 10

2.7 Principles for Choosing Coordinates 11

3 General Change of Variables in $\mathbb{R}^n$ **11**

3.1 The Jacobian Determinant 11

3.2 How the Special Coordinate Systems Fit the Formula 11

3.3 General Integrals in Dimension n 12

3.4 Geometry Applications 12

3.5 Physics Applications 17

4 Practice Problems **20**

5 Summary **20**

Lecture Plan

Lecture 8 introduced double and triple integrals as accumulation over regions. Lecture 9 organizes the computational theory:

- 1. Review double integrals and keep the main techniques for computation.

2. Study triple integrals and how to reduce them to iterated integrals; then choose Cartesian, cylindrical, or spherical coordinates according to the shape of the solid.
3. State the general change-of-variables formula in $\mathbb{R}^n$ and connect it to geometry and physics.

One Guiding Idea

An integral is a sum of many small contributions:

$$\text{density} \times \text{small size}.$$

The only thing that changes from one dimension to higher dimensions is the meaning of “small size”:

$$dx, \quad dA, \quad dV, \quad dx_1 \cdots dx_n.$$

1 Review of Double Integrals from Lecture 8

1.1 Meaning of a Double Integral

For a function $f(x, y)$ on a planar region D , the double integral

$$\iint_D f(x, y) \, dA$$

is the limit of sums

$$\sum f(x_i^*, y_i^*) \Delta A_i.$$

If $f \geq 0$, this is the volume under the surface $z = f(x, y)$ over D . If f is a density, it is mass. If $f = 1$, it is area:

$$\iint_D 1 \, dA = \text{Area}(D).$$

1.2 Computing on Rectangles

For a rectangle

$$R = [a, b] \times [c, d],$$

Fubini’s theorem gives

$$\iint_R f(x, y) \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy,$$

when f is continuous on R .

How to Read an Iterated Integral

Compute from inside to outside. In

$$\int_a^b \int_c^d f(x, y) \, dy \, dx,$$

the inner integral is with respect to y , so x is treated as a constant during the first step.

Example. Compute

$$\iint_{[0,2] \times [1,3]} (x^2 + y) \, dA.$$

Use the order $dy \, dx$:

$$\int_0^2 \int_1^3 (x^2 + y) \, dy \, dx = \int_0^2 \left[x^2 y + \frac{y^2}{2} \right]_1^3 \, dx.$$

The inner integral is $2x^2 + 4$, so

$$\int_0^2 (2x^2 + 4) \, dx = \left[\frac{2x^3}{3} + 4x \right]_0^2 = \frac{40}{3}.$$

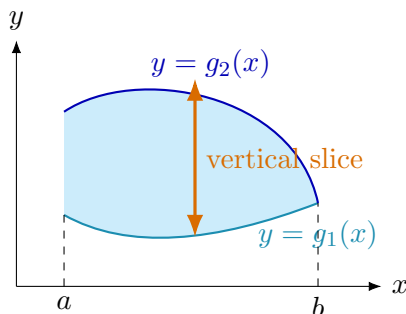
1.3 Type I and Type II Regions

A Type I region is described by vertical slices:

$$D = \{(x, y) : a \leq x \leq b, \, g_1(x) \leq y \leq g_2(x)\}.$$

Then

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$



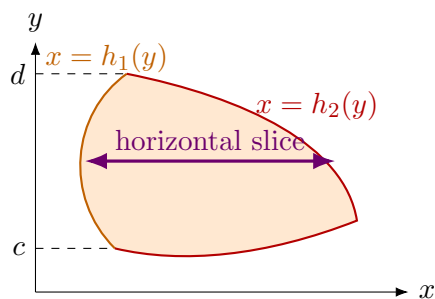
Type I: choose x first, then $g_1(x) \leq y \leq g_2(x)$

A Type II region is described by horizontal slices:

$$D = \{(x, y) : c \leq y \leq d, \, h_1(y) \leq x \leq h_2(y)\}.$$

Then

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy.$$



Type II: choose y first, then $h_1(y) \leq x \leq h_2(y)$

Condition on the Region

The chosen slices should hit the region in one interval, except possibly at boundary points. If a slice enters and leaves the region several times, split the region into simpler pieces.

Example. Let D be the triangle bounded by

$$x = 0, \quad y = 0, \quad x + y = 1.$$

Then

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}.$$

Thus

$$\iint_D (x + y) \, dA = \int_0^1 \int_0^{1-x} (x + y) \, dy \, dx.$$

The value is

$$\int_0^1 \left(x(1-x) + \frac{(1-x)^2}{2} \right) dx = \int_0^1 \left(\frac{1}{2} - \frac{x^2}{2} \right) dx = \frac{1}{3}.$$

1.4 Changing the Order of Integration

Changing the order is a geometric operation:

1. Translate the old bounds into inequalities describing D .
2. Draw or mentally identify the region.
3. Rewrite the same region using slices in the new direction.
4. Check that every point is counted once.

Example. Rewrite

$$\int_0^1 \int_y^1 f(x, y) \, dx \, dy$$

in the order $dy \, dx$. The original bounds say

$$0 \leq y \leq 1, \quad y \leq x \leq 1,$$

so the region is $0 \leq y \leq x \leq 1$. Hence

$$0 \leq x \leq 1, \quad 0 \leq y \leq x,$$

and

$$\int_0^1 \int_y^1 f(x, y) \, dx \, dy = \int_0^1 \int_0^x f(x, y) \, dy \, dx.$$

Why this matters. The integral

$$\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$$

is difficult in the given order because e^{y^2} has no elementary antiderivative in y . The region is

$$0 \leq x \leq y \leq 1.$$

Changing order gives

$$\int_0^1 \int_0^y e^{y^2} \, dx \, dy = \int_0^1 y e^{y^2} \, dy = \frac{e-1}{2}.$$

1.5 Polar Coordinates for Double Integrals

Polar coordinates are

$$x = r \cos \theta, \quad y = r \sin \theta.$$

The area element is not $dr \, d\theta$; it is

$$dA = r \, dr \, d\theta.$$

Therefore

$$\iint_D f(x, y) \, dA = \iint_{D^*} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta.$$

When Polar Coordinates Are Natural

Use polar coordinates when the region or integrand contains

$$x^2 + y^2, \quad \sqrt{x^2 + y^2}, \quad \text{circles, disks, annuli, sectors, or rays.}$$

The factor r is the area-scaling factor of the coordinate map.

Example. For the disk $x^2 + y^2 \leq R^2$,

$$0 \leq r \leq R, \quad 0 \leq \theta \leq 2\pi.$$

The area is

$$\iint_D 1 \, dA = \int_0^{2\pi} \int_0^R r \, dr \, d\theta = \pi R^2.$$

1.6 Double-Integral Technique Checklist

Checklist

1. Identify whether the region is rectangular, Type I, Type II, or polar.
2. Choose the order that gives the simpler bounds or easier antiderivative.
3. If one order gives a non-elementary inner integral, try changing the order.
4. If circles or $x^2 + y^2$ dominate, try polar coordinates.
5. Always include the correct area element: $dA = dx \, dy$ in Cartesian and $dA = r \, dr \, d\theta$ in polar coordinates.

2 Triple Integrals and Coordinate Systems

2.1 Meaning of a Triple Integral

For a function $f(x, y, z)$ on a solid region $E \subset \mathbb{R}^3$, the triple integral

$$\iiint_E f(x, y, z) \, dV$$

is the limit of sums

$$\sum f(x_i^*, y_i^*, z_i^*) \Delta V_i.$$

If $f = 1$, it gives volume:

$$\iiint_E 1 \, dV = \text{Vol}(E).$$

If f is mass density, it gives mass:

$$M = \iiint_E \rho(x, y, z) \, dV.$$

2.2 Rectangular Boxes and Fubini's Theorem

For a box

$$B = [a, b] \times [c, d] \times [p, q],$$

Fubini's theorem gives

$$\iiint_B f(x, y, z) \, dV = \int_a^b \int_c^d \int_p^q f(x, y, z) \, dz \, dy \, dx,$$

and any other order is allowed if f is continuous.

2.3 Reduction to a Double Integral and a Single Integral

Many solids are described by a base region in a coordinate plane plus two bounding surfaces.

Vertical Solids

Suppose

$$E = \{(x, y, z) : (x, y) \in D, u(x, y) \leq z \leq v(x, y)\}.$$

Then

$$\iiint_E f(x, y, z) \, dV = \iint_D \left(\int_{u(x, y)}^{v(x, y)} f(x, y, z) \, dz \right) dA.$$

If $f = 1$, this becomes the height formula

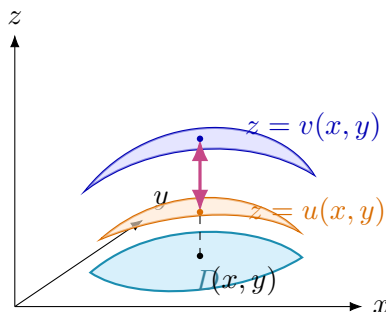
$$\text{Vol}(E) = \iint_D (v(x, y) - u(x, y)) \, dA.$$

Condition for a Vertical Description

Every vertical line through a point (x, y) in the base D must intersect the solid in one interval

$$u(x, y) \leq z \leq v(x, y).$$

If a vertical line cuts the solid in several separated pieces, split the solid.



A vertical fiber over each point of D should be one interval in z .

Example: Under a Paraboloid. Let E be the solid under

$$z = 4 - x^2 - y^2$$

and above the xy -plane. The base is the disk

$$D = \{(x, y) : x^2 + y^2 \leq 4\}.$$

Thus

$$\text{Vol}(E) = \iint_D (4 - x^2 - y^2) \, dA.$$

Using polar coordinates in the base,

$$\text{Vol}(E) = \int_0^{2\pi} \int_0^2 (4 - r^2)r \, dr \, d\theta = 8\pi.$$

Horizontal Slices

Sometimes it is better to slice by levels of z . If

$$E = \{(x, y, z) : \alpha \leq z \leq \beta, (x, y) \in D_z\},$$

then

$$\iiint_E f(x, y, z) \, dV = \int_\alpha^\beta \left(\iint_{D_z} f(x, y, z) \, dA \right) dz.$$

This is a single integral of cross-sectional double integrals.

Example: Sphere by Cross Sections. For the ball

$$x^2 + y^2 + z^2 \leq R^2,$$

the slice at height z is a disk

$$x^2 + y^2 \leq R^2 - z^2.$$

Hence

$$\text{Vol}(B_R) = \int_{-R}^R \text{Area}(D_z) \, dz = \int_{-R}^R \pi(R^2 - z^2) \, dz = \frac{4}{3}\pi R^3.$$

Other Simple Directions

The same idea works if x or y is the inner variable:

$$E = \{(x, y, z) : (y, z) \in G, a(y, z) \leq x \leq b(y, z)\}$$

gives

$$\iiint_E f \, dV = \iint_G \left(\int_{a(y,z)}^{b(y,z)} f(x, y, z) \, dx \right) dA_{yz}.$$

Choose the direction in which the solid is easiest to describe.

2.4 Cartesian Bounds for Common Solids

Tetrahedron. Let E be bounded by the coordinate planes and

$$x + y + z = 1.$$

Then

$$E = \{(x, y, z) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x, 0 \leq z \leq 1 - x - y\}.$$

Therefore

$$\text{Vol}(E) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1 \, dz \, dy \, dx = \frac{1}{6}.$$

The Hard Part

For triple integrals, most mistakes happen before integration begins: wrong projection, wrong lower/upper surface, missing split of the region, or using the wrong volume element.

2.5 Cylindrical Coordinates

Cylindrical coordinates are polar coordinates in the xy -plane with the height variable z :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

The volume element is

$$dV = r \, dz \, dr \, d\theta,$$

or the same factor r with any order of dz , dr , and $d\theta$.

How the Volume Changes

A small rectangular box in (r, θ, z) -space has side lengths

$$\Delta r, \quad \Delta \theta, \quad \Delta z.$$

But after the map

$$T(r, \theta, z) = (r \cos \theta, r \sin \theta, z),$$

the three coordinate directions do not keep the same physical lengths.

First fix z and look only in the xy -plane. Increasing r by Δr moves radially, so the radial side has length approximately

$$\Delta r.$$

Increasing θ by $\Delta\theta$ moves along a circular arc of radius r , so the angular side has length approximately

$$\text{arc length} = r\Delta\theta.$$

Increasing z by Δz moves vertically, so the vertical side has length

$$\Delta z.$$

Therefore the small physical volume is approximately

$$\Delta r (r\Delta\theta) \Delta z = r \Delta r \Delta\theta \Delta z.$$

The same result comes from the Jacobian determinant. The Jacobian matrix is

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Its determinant is

$$\begin{aligned} \det \frac{\partial(x, y, z)}{\partial(r, \theta, z)} &= 1 \cdot \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} \\ &= r \cos^2 \theta + r \sin^2 \theta \\ &= r. \end{aligned}$$

Thus a tiny coordinate volume $\Delta r \Delta\theta \Delta z$ becomes approximately

$$|\det J| \Delta r \Delta\theta \Delta z = r \Delta r \Delta\theta \Delta z.$$

That is why the volume element is

$$\boxed{dV = r \, dr \, d\theta \, dz}.$$

When Cylindrical Coordinates Are Natural

Use cylindrical coordinates when the solid has circular symmetry around the z -axis, or when the equations contain

$$x^2 + y^2, \quad r, \quad z.$$

Typical examples: cylinders, cones, paraboloids, circular disks extruded in the z direction.

Example: Cylinder with Variable Height. Find the volume of

$$x^2 + y^2 \leq 9, \quad 0 \leq z \leq 4 - y.$$

In cylindrical coordinates, $y = r \sin \theta$, so

$$0 \leq r \leq 3, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq 4 - r \sin \theta.$$

Thus

$$V = \int_0^{2\pi} \int_0^3 \int_0^{4-r\sin\theta} r \, dz \, dr \, d\theta.$$

The inner integral gives

$$V = \int_0^{2\pi} \int_0^3 (4 - r \sin \theta) r \, dr \, d\theta.$$

Since $\int_0^{2\pi} \sin \theta \, d\theta = 0$,

$$V = \int_0^{2\pi} \int_0^3 4r \, dr \, d\theta = 36\pi.$$

2.6 Spherical Coordinates

Spherical coordinates are

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

Here ρ is distance from the origin, θ is the angle in the xy -plane, and ϕ is the angle down from the positive z -axis.

The volume element is

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

How the Volume Changes

In spherical coordinates, a small coordinate box has approximate side lengths

$$\Delta\rho, \quad \rho \Delta\phi, \quad \rho \sin \phi \Delta\theta.$$

Multiplying these gives

$$\Delta V \approx \rho^2 \sin \phi \Delta\rho \Delta\phi \Delta\theta.$$

Therefore

$$\boxed{dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta}.$$

When Spherical Coordinates Are Natural

Use spherical coordinates when the solid or integrand depends on

$$x^2 + y^2 + z^2 = \rho^2$$

or when the boundary is a sphere, cone through the origin, or ball-shaped shell. Cones through the origin usually become constant ϕ surfaces.

Example: Ball. For the ball $x^2 + y^2 + z^2 \leq R^2$,

$$0 \leq \rho \leq R, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

Thus

$$\text{Vol}(B_R) = \int_0^{2\pi} \int_0^\pi \int_0^R \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{4}{3}\pi R^3.$$

Example: Cone and Sphere. Consider the solid inside $\rho \leq 2$ and above the cone $\phi = \pi/4$. Then

$$0 \leq \rho \leq 2, \quad 0 \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \theta \leq 2\pi.$$

The volume is

$$\int_0^{2\pi} \int_0^{\pi/4} \int_0^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{16\pi}{3} \left(1 - \frac{\sqrt{2}}{2}\right).$$

2.7 Principles for Choosing Coordinates

Coordinate system	Best signs	Volume/area element
Cartesian	Planes, boxes, simple graph bounds	$dx \, dy \, dz$
Polar	Planar disks, annuli, sectors	$r \, dr \, d\theta$
Cylindrical	Cylinders, cones, paraboloids around z -axis	$r \, dr \, d\theta \, dz$
Spherical	Balls, spherical shells, cones from origin	$\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$

Coordinate Choice Principle

Choose coordinates so that the boundary equations and the integrand become simpler at the same time. A good coordinate system usually turns curved boundaries into constant bounds and turns complicated expressions into one coordinate variable.

3 General Change of Variables in $\mathbb{R}^n$

3.1 The Jacobian Determinant

Let

$$T(u_1, \dots, u_n) = (x_1(u), \dots, x_n(u))$$

be a smooth one-to-one coordinate change from a region S in u -space to a region D in x -space. The Jacobian matrix is

$$DT = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \cdots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}.$$

Its determinant measures local volume scaling.

Theorem 3.1 (Change of Variables in $\mathbb{R}^n$). *If $T : S \rightarrow D$ is a smooth one-to-one change of variables with nonzero Jacobian determinant except possibly on boundary pieces, then*

$$\int_D f(x) \, dx_1 \cdots dx_n = \int_S f(T(u)) \, |\det DT(u)| \, du_1 \cdots du_n.$$

Why the Absolute Value Appears

The determinant can be negative if the coordinate map reverses orientation. Integrals over size use positive area or volume, so the scaling factor is $|\det DT|$.

3.2 How the Special Coordinate Systems Fit the Formula

For polar coordinates,

$$T(r, \theta) = (r \cos \theta, r \sin \theta),$$

so

$$\det \frac{\partial(x, y)}{\partial(r, \theta)} = r.$$

Thus $dA = r dr d\theta$.

For cylindrical coordinates,

$$T(r, \theta, z) = (r \cos \theta, r \sin \theta, z),$$

so

$$|\det DT| = r.$$

Thus $dV = r dr d\theta dz$.

For spherical coordinates,

$$T(\rho, \phi, \theta) = (\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi),$$

so

$$|\det DT| = \rho^2 \sin \phi.$$

Thus $dV = \rho^2 \sin \phi d\rho d\phi d\theta$.

3.3 General Integrals in Dimension n

For a region $D \subset \mathbb{R}^n$,

$$\int_D f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

means accumulation over n -dimensional size. If $f = 1$, it gives n -dimensional volume:

$$\int_D 1 dx_1 \cdots dx_n = \text{Vol}_n(D).$$

If D is a product box

$$D = [a_1, b_1] \times \cdots \times [a_n, b_n],$$

then Fubini's theorem allows repeated one-variable integration in any order:

$$\int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \cdots dx_1.$$

3.4 Geometry Applications

Area and Volume. Use $f = 1$:

$$\text{Area}(D) = \iint_D 1 dA, \quad \text{Vol}(E) = \iiint_E 1 dV.$$

Volume under a surface. If $z = f(x, y) \geq 0$ over D , then

$$V = \iint_D f(x, y) dA.$$

Volume between surfaces. If

$$u(x, y) \leq z \leq v(x, y),$$

then

$$V = \iint_D (v(x, y) - u(x, y)) dA.$$

The n -dimensional ball and sphere. Let

$$B_R^n = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leq R\}, \quad S_R^{n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = R\}.$$

The ball B_R^n has dimension n . Its boundary S_R^{n-1} has dimension $n - 1$.

Start from low dimensions. In one dimension, the “ball” is just an interval:

$$B_R^1 = [-R, R].$$

So

$$\text{Vol}_1(B_R^1) = \int_{-R}^R 1 \, dx = 2R.$$

In two dimensions, B_R^2 is a disk. Slice the disk horizontally. At height y , the slice is an interval

$$-\sqrt{R^2 - y^2} \leq x \leq \sqrt{R^2 - y^2},$$

so its length is $2\sqrt{R^2 - y^2}$. Therefore

$$\text{Vol}_2(B_R^2) = \int_{-R}^R 2\sqrt{R^2 - y^2} \, dy.$$

This integral is the area of a semicircle of radius R , doubled, so

$$\text{Vol}_2(B_R^2) = \pi R^2.$$

In three dimensions, B_R^3 is the usual solid ball. Slice it by horizontal planes $z = \text{constant}$. At height z , the cross-section is a disk of radius

$$\sqrt{R^2 - z^2}.$$

Using z as the slicing variable,

$$\text{Vol}_3(B_R^3) = \int_{-R}^R \pi(R^2 - z^2) \, dz.$$

Compute directly:

$$\text{Vol}_3(B_R^3) = \pi \left[R^2 z - \frac{z^3}{3} \right]_{-R}^R = \frac{4}{3} \pi R^3.$$

Guess the general pattern. The previous computations all have the same structure:

$$\text{volume} = \int \text{volume of cross-section} \, d(\text{slicing variable}).$$

For the n -dimensional ball, write the last coordinate as $t = x_n$. At height t , the cross-section is an $(n - 1)$ -dimensional ball:

$$B_{\sqrt{R^2 - t^2}}^{n-1}.$$

Thus the natural recursion is

$$\text{Vol}_n(B_R^n) = \int_{-R}^R \text{Vol}_{n-1} \left(B_{\sqrt{R^2 - t^2}}^{n-1} \right) \, dt.$$

General Recursion

Suppose

$$\text{Vol}_{n-1}(B_s^{n-1}) = C_{n-1}s^{n-1},$$

where

$$C_{n-1} = \text{Vol}_{n-1}(B_1^{n-1})$$

is the volume of the unit $(n-1)$ -ball. Then

$$\text{Vol}_n(B_R^n) = C_{n-1} \int_{-R}^R (R^2 - t^2)^{(n-1)/2} dt.$$

If

$$t = R \sin \theta,$$

then

$$dt = R \cos \theta \, d\theta, \quad \sqrt{R^2 - t^2} = R \cos \theta.$$

Then

$$\text{Vol}_n(B_R^n) = C_{n-1} R^n \int_{-\pi/2}^{\pi/2} \cos^n \theta \, d\theta.$$

Define

$$I_n = \int_{-\pi/2}^{\pi/2} \cos^n \theta \, d\theta.$$

By integration by parts,

$$I_n = \frac{n-1}{n} I_{n-2}, \quad n \geq 2,$$

with

$$I_0 = \pi, \quad I_1 = 2.$$

Thus the constants $C_n = \text{Vol}_n(B_1^n)$ satisfy

$$C_n = C_{n-1} I_n.$$

This gives

$$\text{Vol}_n(B_R^n) = C_n R^n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} R^n.$$

Useful Special Cases

$$\text{Vol}_1(B_R^1) = 2R, \quad \text{Vol}_2(B_R^2) = \pi R^2, \quad \text{Vol}_3(B_R^3) = \frac{4}{3} \pi R^3.$$

$$\text{Vol}_4(B_R^4) = \frac{\pi^2}{2} R^4, \quad \text{Vol}_5(B_R^5) = \frac{8\pi^2}{15} R^5.$$

Surface area of a higher-dimensional sphere. The sphere S_R^{n-1} is the boundary of the ball B_R^n . A thin shell of thickness ΔR has volume approximately

$$\text{Area}(S_R^{n-1}) \Delta R.$$

Therefore calculus gives

$$\text{Area}(S_R^{n-1}) = \frac{d}{dR} \text{Vol}_n(B_R^n).$$

Since

$$\text{Vol}_n(B_R^n) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} R^n,$$

we get

$$\text{Area}(S_R^{n-1}) = \frac{n\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} R^{n-1}.$$

Using

$$\Gamma\left(\frac{n}{2} + 1\right) = \frac{n}{2} \Gamma\left(\frac{n}{2}\right),$$

this becomes

$$\text{Area}(S_R^{n-1}) = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})} R^{n-1}.$$

Quick Checks

For $n = 2$, the sphere S_R^1 is a circle:

$$\text{Area}(S_R^1) = 2\pi R.$$

For $n = 3$, the sphere S_R^2 is the usual sphere:

$$\text{Area}(S_R^2) = 4\pi R^2.$$

For $n = 4$, the sphere S_R^3 has three-dimensional surface area:

$$\text{Area}(S_R^3) = 2\pi^2 R^3.$$

Average value. For a finite region D ,

$$f_{\text{avg}} = \frac{1}{\text{Vol}_n(D)} \int_D f \, dV_n.$$

Average of a harmonic function on a higher-dimensional sphere. Let

$$S_R^{n-1}(\mathbf{a}) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| = R\}$$

be the sphere of radius R centered at $\mathbf{a}$. Its dimension is $n - 1$ because it is the boundary of an n -dimensional ball.

Mean Value Property for Harmonic Functions

If u is harmonic on a ball containing $S_R^{n-1}(\mathbf{a})$ and its interior, meaning

$$\Delta u = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_n^2} = 0,$$

then the average of u over the sphere equals the value at the center:

$$\frac{1}{\text{Area}(S_R^{n-1}(\mathbf{a}))} \int_{S_R^{n-1}(\mathbf{a})} u(\mathbf{x}) \, dS = u(\mathbf{a}).$$

Example

In $\mathbb{R}^n$ with $n \geq 3$, consider

$$u(x_1, \dots, x_n) = x_1^2 - x_2^2 + 3x_3 + 5.$$

Since

$$\Delta u = 2 - 2 + 0 + \cdots + 0 = 0,$$

the function is harmonic. Therefore its average over the sphere

$$S_4^{n-1}(\mathbf{a}), \quad \mathbf{a} = (1, 2, 0, \dots, 0),$$

is simply

$$u(\mathbf{a}) = 1^2 - 2^2 + 3 \cdot 0 + 5 = 2.$$

So

$$\frac{1}{\text{Area}(S_4^{n-1}(\mathbf{a}))} \int_{S_4^{n-1}(\mathbf{a})} (x_1^2 - x_2^2 + 3x_3 + 5) \, dS = 2.$$

This example shows a useful geometric idea: for harmonic functions, the center value controls the spherical average in every dimension. In practice, this can replace a difficult surface integral over a high-dimensional sphere with one function evaluation.

Why the example also fits symmetry. Write a point on the sphere as

$$\mathbf{x} = \mathbf{a} + R\xi, \quad \|\xi\| = 1.$$

By symmetry on the unit sphere,

$$\text{avg}(\xi_i) = 0, \quad \text{avg}(\xi_i^2) = \frac{1}{n}.$$

Thus

$$\text{avg}(x_1^2) = a_1^2 + \frac{R^2}{n}, \quad \text{avg}(x_2^2) = a_2^2 + \frac{R^2}{n},$$

so the quadratic parts cancel:

$$\text{avg}(x_1^2 - x_2^2 + 3x_3 + 5) = a_1^2 - a_2^2 + 3a_3 + 5 = u(\mathbf{a}).$$

3.5 Physics Applications

Mass. If ρ is density, then

$$M = \int_D \rho \, dV_n.$$

In two dimensions, this is lamina mass:

$$M = \iint_D \rho(x, y) \, dA.$$

In three dimensions, it is solid mass:

$$M = \iiint_E \rho(x, y, z) \, dV.$$

Center of mass from torque. The center of mass is the point where the object balances. A useful way to express this idea is through torque.

Let a point in the solid be

$$\mathbf{r} = (x, y, z),$$

and let the unknown center of mass be

$$\mathbf{R} = (\bar{x}, \bar{y}, \bar{z}).$$

In a uniform gravitational field $\mathbf{g}$, the small force on a small mass element is

$$d\mathbf{F} = \mathbf{g} \, dm, \quad dm = \rho(x, y, z) \, dV.$$

The torque about the point $\mathbf{R}$ is

$$d\boldsymbol{\tau} = (\mathbf{r} - \mathbf{R}) \times d\mathbf{F} = (\mathbf{r} - \mathbf{R}) \times \mathbf{g} \, dm.$$

The total torque is therefore

$$\boldsymbol{\tau}(\mathbf{R}) = \iiint_E (\mathbf{r} - \mathbf{R}) \times \mathbf{g} \, dm.$$

Balance Definition

The center of mass is the point $\mathbf{R}$ for which the total first moment vector vanishes:

$$\iiint_E (\mathbf{r} - \mathbf{R}) \, dm = \mathbf{0}.$$

For every uniform force direction, this makes the total torque about $\mathbf{R}$ equal to zero.

Indeed, since $\mathbf{g}$ is constant,

$$\boldsymbol{\tau}(\mathbf{R}) = \left(\iiint_E (\mathbf{r} - \mathbf{R}) \, dm \right) \times \mathbf{g}.$$

So the stronger balance condition

$$\iiint_E (\mathbf{r} - \mathbf{R}) \, dm = \mathbf{0}$$

implies

$$\boldsymbol{\tau}(\mathbf{R}) = \mathbf{0}.$$

A Small Caveat

If the direction of gravity is fixed, zero torque can hold for any point on the vertical line through the center of mass. To get the unique point $\mathbf{R}$, use the vector condition

$$\iiint_E (\mathbf{r} - \mathbf{R}) \, dm = \mathbf{0},$$

or require zero torque for every uniform force direction.

Deriving the coordinate formulas. Expand the balance condition:

$$\iiint_E (\mathbf{r} - \mathbf{R}) \, dm = \mathbf{0}.$$

Since $\mathbf{R}$ is constant,

$$\iiint_E \mathbf{r} \, dm - \mathbf{R} \iiint_E dm = \mathbf{0}.$$

The total mass is

$$M = \iiint_E dm = \iiint_E \rho(x, y, z) \, dV.$$

Therefore

$$\mathbf{R} = \frac{1}{M} \iiint_E \mathbf{r} \, dm.$$

In coordinates,

$$\bar{x} = \frac{1}{M} \iiint_E x \rho \, dV, \quad \bar{y} = \frac{1}{M} \iiint_E y \rho \, dV, \quad \bar{z} = \frac{1}{M} \iiint_E z \rho \, dV.$$

First moments in three dimensions. The quantities

$$M_{yz} = \iiint_E x \rho \, dV, \quad M_{xz} = \iiint_E y \rho \, dV, \quad M_{xy} = \iiint_E z \rho \, dV$$

are called the first moments of mass. They measure how much mass is distributed away from the coordinate planes:

M_{yz} is the moment about the yz -plane,

M_{xz} is the moment about the xz -plane, M_{xy} is the moment about the xy -plane.

The center-of-mass formulas become

$$\bar{x} = \frac{M_{yz}}{M}, \quad \bar{y} = \frac{M_{xz}}{M}, \quad \bar{z} = \frac{M_{xy}}{M}.$$

Center of Mass Computation

1. Compute the total mass:

$$M = \iiint_E \rho(x, y, z) \, dV.$$

2. Compute the first moments:

$$M_{yz} = \iiint_E x\rho \, dV, \quad M_{xz} = \iiint_E y\rho \, dV, \quad M_{xy} = \iiint_E z\rho \, dV.$$

3. Divide each moment by the total mass:

$$\bar{x} = \frac{M_{yz}}{M}, \quad \bar{y} = \frac{M_{xz}}{M}, \quad \bar{z} = \frac{M_{xy}}{M}.$$

For a lamina D in the plane,

$$M = \iint_D \rho(x, y) \, dA,$$

and

$$\bar{x} = \frac{1}{M} \iint_D x\rho(x, y) \, dA, \quad \bar{y} = \frac{1}{M} \iint_D y\rho(x, y) \, dA.$$

Symmetry shortcut. If both the region and the density are symmetric with respect to a coordinate plane, the corresponding coordinate of the center of mass lies on that plane. For example, if a solid and its density are symmetric with respect to the plane $x = 0$, then $\bar{x} = 0$.

Moment of inertia. The moment of inertia around the z -axis is

$$I_z = \iiint_E (x^2 + y^2)\rho(x, y, z) \, dV.$$

In cylindrical coordinates this becomes

$$I_z = \iiint_{E^*} r^2 \rho(r \cos \theta, r \sin \theta, z) r \, dr \, d\theta \, dz.$$

The extra r is still the Jacobian factor.

Probability. If $p(x_1, \dots, x_n)$ is a probability density on D , then

$$\int_D p(x_1, \dots, x_n) \, dx_1 \cdots dx_n = 1.$$

The probability of an event region $A \subset D$ is

$$P(A) = \int_A p(x_1, \dots, x_n) \, dx_1 \cdots dx_n.$$

4 Practice Problems

1. Change the order of integration:

$$\int_0^1 \int_x^1 f(x, y) \, dy \, dx.$$

2. Compute

$$\iint_{x^2+y^2 \leq 4} (x^2 + y^2) \, dA.$$

3. Set up the volume of the solid below $z = 9 - x^2 - y^2$ and above $z = 0$ using Cartesian bounds and then cylindrical bounds.
4. Compute the volume of the tetrahedron bounded by $x = 0$, $y = 0$, $z = 0$, and $x + 2y + z = 4$.
5. Change the order of the triple iterated integral. Rewrite

$$\int_{-1}^1 \int_{x^2}^1 \int_0^{1-y} f(x, y, z) \, dz \, dy \, dx$$

in the order $dx \, dy \, dz$.

6. Describe the ball $x^2 + y^2 + z^2 \leq 9$ in spherical coordinates and compute its volume.
7. Decide which coordinate system is best and explain why:

$$x^2 + y^2 \leq z \leq 4.$$

8. Find the mass of the cylinder $x^2 + y^2 \leq 1$, $0 \leq z \leq 2$, with density $\rho(x, y, z) = 1 + x^2 + y^2$.
9. Use the general change-of-variables formula for

$$x = 2u, \quad y = 3v$$

to compute the area of the ellipse

$$\frac{x^2}{4} + \frac{y^2}{9} \leq 1.$$

5 Summary

- Double integrals are computed by choosing good slices or good coordinates.
- Triple integrals can often be reduced to a double integral over a base and a single integral through the solid.
- Region conditions matter: a chosen slicing direction should intersect the region in one interval, or the region must be split.
- Cylindrical coordinates use $dV = r \, dr \, d\theta \, dz$.
- Spherical coordinates use $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.
- The general change-of-variables formula multiplies by $|\det DT|$.
- Geometry and physics applications come from choosing the integrand: 1 for size, density for mass, distance-squared times density for moment of inertia.